

**111EQUATION CHAPTER 1 SECTION 1
BRIDGING THE GAP: CONNECTING UNDERSERVED LOS
ANGELES NEIGHBOURHOODS TO BRT**

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Abstract

This report examines how targeted feeder bus routes can strengthen access to Los Angeles’ Bus Rapid Transit (BRT) corridors, addressing persistent mobility gaps in dense, low-income, and highly carless neighbourhoods. Using spatial and demographic analysis, the study identifies communities located beyond walking distance of G and J Line stations yet disproportionately reliant on public transport. Building on international evidence from LMIC example, Bogotá, and local example, Houston, the report evaluates how integrating short, frequent feeder routes with existing trunk services can improve accessibility, reliability, and equity. A phased implementation plan is proposed, detailing technical feasibility, community engagement, fleet strategy, and operational considerations. The analysis shows that feeder-route deployment is financially and institutionally viable, particularly when combined with LA Metro’s transition toward a zero-emission bus fleet. In doing so, the project demonstrates how relatively low-cost, context-sensitive interventions can reduce spatial inequalities, increase utilisation of the BRT network, and support long-term sustainability goals for Los Angeles’ transit system.

Keywords: Engineering, International Development, Public Transit, Bus Rapid Transport, Feeder Routes

1. INTRODUCTION

Los Angeles is widely characterised as a car-oriented metropolis, yet buses remain the backbone of its public transport system, accounting for more than 70% of all transit trips. Despite this reliance, public transport performance has declined in several districts over the past decade. Rising living costs and a dispersed urban form have shifted lower-income households away from central neighbourhoods into areas with weaker transport provision, increasing dependence on local bus services. In response, LA Metro has invested in higher-quality bus rapid transit (BRT), most notably the G Line and J Line, to improve speed and reliability along major corridors. While these investments represent an important step forward, they also expose a structural limitation in how benefits are distributed across the wider network.

Access to BRT corridors remains highly uneven. Many lower-income neighbourhoods lie beyond walking distance of stations and continue to rely on slower local routes with limited connectivity. In a city where car ownership is not evenly distributed, weak links between local districts and the BRT backbone restrict access to employment, education, and essential services. The NextGen Bus Plan, introduced in 2020, sought to modernise the bus system by increasing frequency and simplifying schedules. However, it did not explicitly reconfigure local routes to function as feeders to BRT. Thus, leaving accessibility gaps in dense, low-income, and car-poor communities that remain heavily dependent on bus travel.

These conditions provide a clear justification for intervention in the context of engineering and international development. The Los Angeles context is examined as a case study to explore how principles commonly applied in low and middle-income countries, such as Bogata, can be applied within high-income cities that contain underserved neighbourhoods. Example principles explored from this include maximising network coverage and delivering cost-effective improvements through incremental interventions. This presents itself as another means of justifications for Los Angeles being our primary focus. This approach reflects a central concern of engineering and international development: addressing inequality through appropriate, context-sensitive infrastructure rather than assuming that higher levels of investment alone resolve access deficits. Although Los Angeles is a high-income context, the spatial inequalities revealed here mirror challenges typically examined in international development: unequal access to essential services, infrastructural exclusion, and institutional fragmentation

Against this backdrop, the objective of this project is twofold:

1. To identify which neighbourhoods experience the greatest accessibility gaps to BRT, using demographic and transit data;
2. To propose a feeder-based intervention to strengthen links between local districts and major BRT corridors.

Using spatial and demographic analysis informed by transport planning and international development practice, the study assesses how targeted engineering solutions can improve network performance while remaining inclusive, economically feasible, and sustainable.

2. EMPIRICAL PATTERNS AND PROBLEM DEFINITION

Using Census/ACS socio-demographic data and LA Metro GTFS data to build empirical maps, we found that, from an international development perspective, the patterns that emerge highlight a classic accessibility deficit: dense, low-income communities face structural barriers to mobility despite proximity to major infrastructure (Durga Niraula, n.d.; BusMaps, n.d.).

Many dense, bottom-30% income census tracts located more than 500 metres from a BRT station

(benchmark for max. walking distance) have no meaningful feeder service: the majority have zero routes connecting them to the G or J Line, and only a small minority have even a single connecting bus.

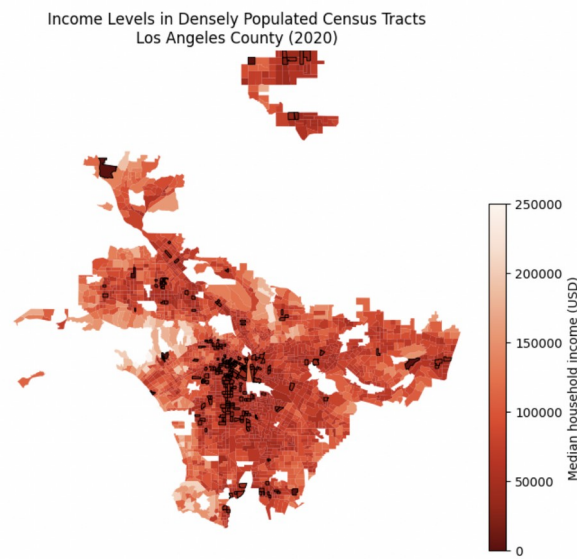


Figure 1. Mapping the median household income by neighbourhood (Niraula, 2024)

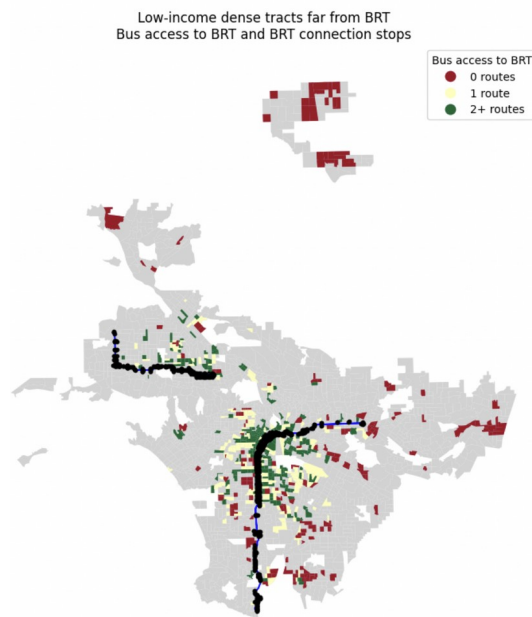


Figure 2. Mapping which lower income households (bottom 30% in terms of median income) that face poor connectivity to BTR stops (Niraula, 2024. And BusMaps, 2025)

Crucially, most of these same tracts also exhibit very high carlessness rates, often exceeding 50–60%, meaning residents cannot substitute poor transit access with private vehicles. As a result, the neighbourhoods with the weakest BRT connectivity are also those most dependent on public transport for daily mobility.

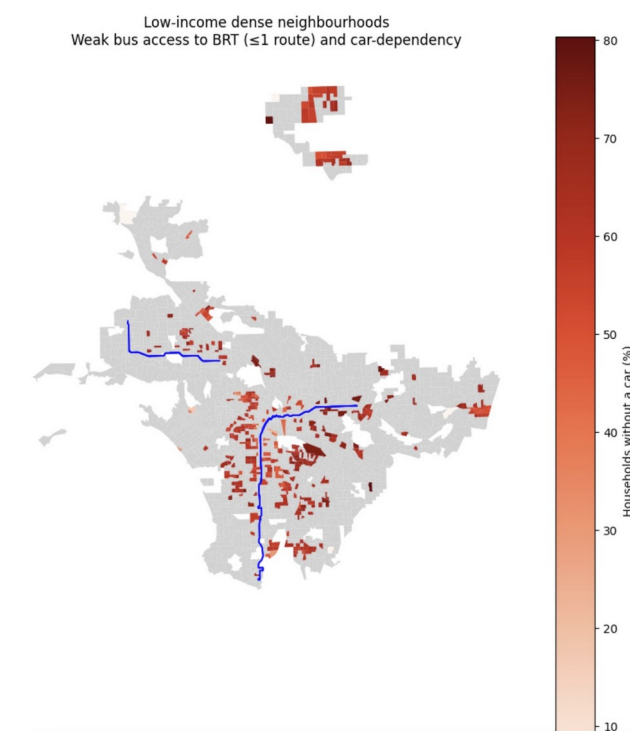


Figure 3. Mapping which lower income households with high carless rates that face poor connectivity to BTR stops (Niraula, 2024. And BusMaps, 2025)

Spatial analysis further shows that these tracts disproportionately overlap with socio-economically disadvantaged communities, notably highly Hispanic dense community areas, where limited access to private vehicles, higher rates of shift work, and constrained housing choices amplify the consequences of poor transit provision.

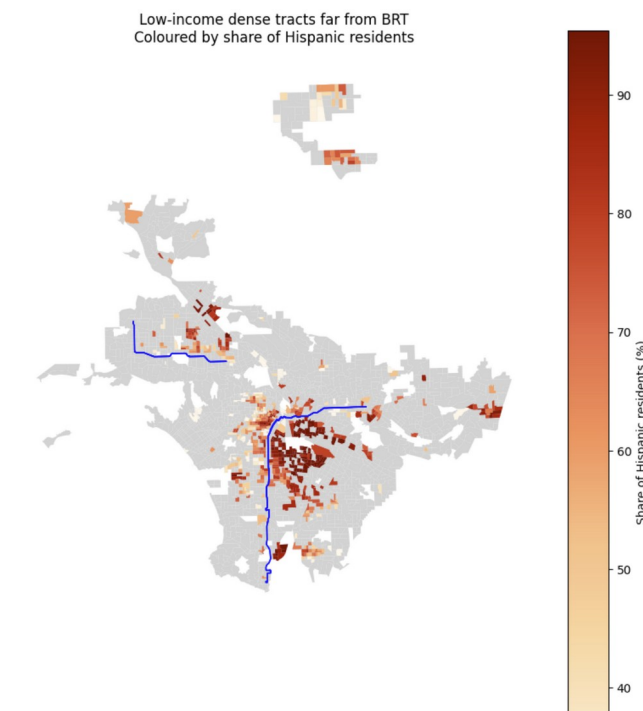


Figure 4. Mapping which of these lower-income households facing poor connectivity to BTR stops and with high carless rates are majorly hispanic. (Niraula, 2024. And BusMaps, 2025)

3. LITERATURE REVIEW

Research across global transit systems consistently shows that integrating high-capacity trunk services with local feeder routes is an effective strategy for improving accessibility, reliability, and ridership. Much of the foundational work on trunk–feeder planning emphasises separating functions: trunk routes provide fast, direct, high-frequency service, while feeders offer localised coverage and deliver passengers to transfer hubs (Kuah and Perl 1987). This structure reduces operational inefficiencies caused by long, circuitous routes and ensures that resources are directed toward corridors where they can achieve the greatest impact.

In practice, the benefits are substantial. Bogotá’s TransMilenio, for example, recorded 40–60% increases in feeder–trunk transfers once coordinated feeders were introduced, illustrating how even modest feeder links can unlock significant ridership gains (Hidalgo et al., n.d.). The system now integrates 112 km of trunk corridors with 440 km of feeder services reaching peripheral and informal settlements (Tiolúwani Aderibigbe et al. 2021). Feeder buses are deliberately short-range and tightly integrated into the fare and operational system, supporting greater equity by enabling residents with limited transport options to access jobs and services. The system has delivered substantial travel time savings and accessibility improvements, though recent analyses highlight the need for adequate investment to sustain quality and avoid overcrowding.



Figure 5. Bogotá’s TransMilenio Bus Fleet (‘TransMilenio - Inicio’, n.d.)

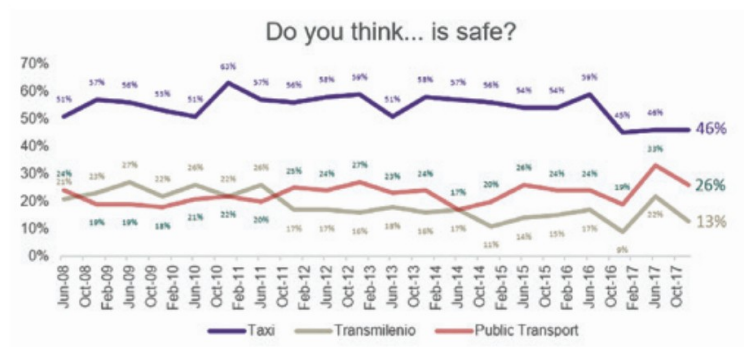


Figure 6. Percentage of surveyed citizens who think it is safe to travel in taxis, TransMilenio or the Public Transit System except the BRT system (Global Infrastructure Hub, n.d.)

In North America, Houston offers a leading example of how redesigning a bus network around frequent, legible corridors improves performance without major cost increases. The 2015 METRO network redesign replaced a radial system with a high-frequency grid, doubling the number of frequent corridors and enabling riders to complete most trips through reliable transfers (NACTO

2016). Within the first year, overall system usage increased, with strong gains on weekends when transfer-based networks are most effective (Leah Binkovitz 2016). Houston demonstrates that reliable transfers, enabled by frequency, are central to any trunk-feeder network.

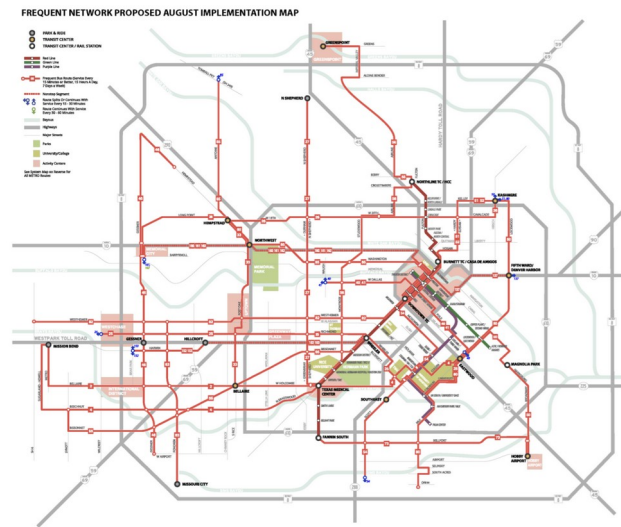


Figure 7. Updated Houston Metro Map showing high-frequency grid system (NACTO 2016)
A consistent theme in these systems is the coverage vs. speed tradeoff. Networks that are designed to maximise ridership concentrate service where demand is strongest, trading some coverage for higher frequency and overall usefulness (Jarrett Walker, n.d.). Academic studies reinforce this tension, showing that systems attempting to maximise geographic coverage often sacrifice reliability and travel speed, particularly for low-income passengers who disproportionately rely on buses (Angurini and Venter 2025).

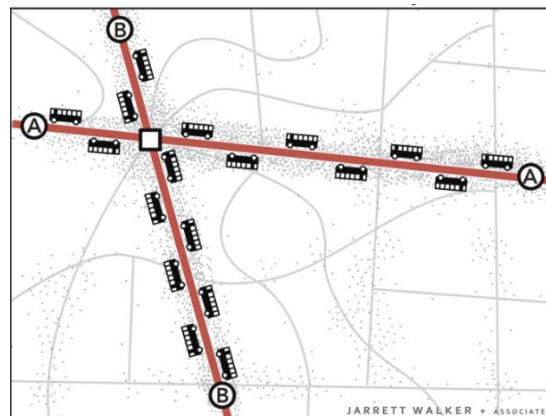


Figure 8. Example map showing designing for maximum ridership (Jarrett Walker, n.d.)

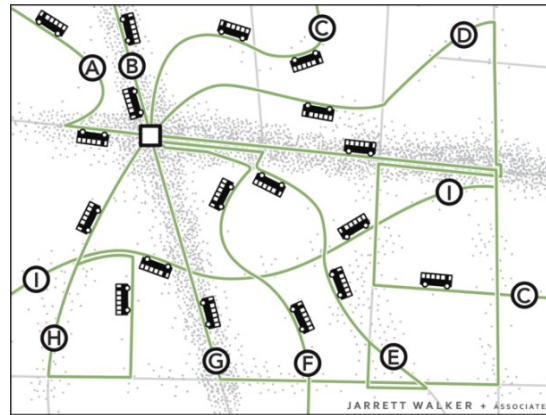


Figure 9. Example map showing designing for maximum coverage (Jarrett Walker, n.d.)

Research on first or last-mile access adds a further dimension. Studies consistently show that feeder networks succeed only when passengers can reliably reach stops and stations on foot. Walkability, sidewalk continuity, street connectivity, perceived safety, and crossing conditions strongly determine whether a feeder-based system is usable (Braun et al. 2022; Kåresdotter et al. 2022; Tokey et al. 2025). In contexts with poor pedestrian infrastructure, the effective catchment area around a trunk station shrinks dramatically, increasing the need for additional feeder or FM/LM services.

Demand-responsive transit (DRT) has been explored as a possible FM/LM solution, but evidence from pilot programmes suggests that DRT is most effective as a complementary service, not a replacement for fixed-route feeders. Studies show that while DRT can fill specific gaps such as very low-density or off-peak areas, it suffers from scalability and cost challenges when used as a primary service (Alonso-González et al. 2018; Currie and Wong 2020). These findings reinforce that the backbone of any successful network must remain high-frequency fixed routes, supported by flexible services.

Within Los Angeles, the NextGen Bus Plan acknowledges many of these principles, aiming to increase frequency and simplify corridors (LA Metro 2020). However, unlike Bogotá, LA lacks a coherent, citywide district-level feeder strategy that systematically links neighbourhoods to rapid bus and rail corridors. While other cities concentrate investment into strong trunks and coordinated feeders, LA's network still distributes service across numerous long routes, reducing frequency and weakening transfer-based connectivity.

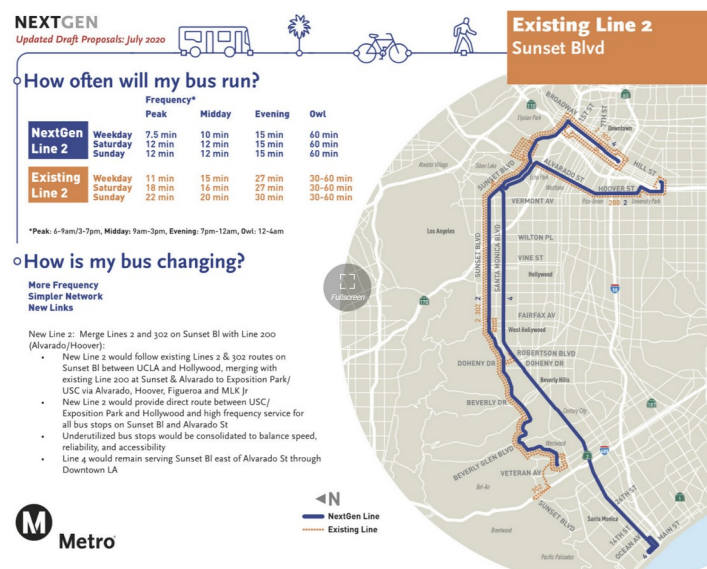


Figure 10. Example Los Angeles' NextGen Bus Plan proposal showing how it focuses on frequency instead of

In conclusion, these comparative experiences suggest a clear gap: Los Angeles has strong trunks but insufficient feeder integration. Building a district-level feeder system would bring LA's network closer to successful international models, improving accessibility, equity, and overall transit performance.

4. PROPOSED PROJECT: STRENGTHENING FEEDER BUS CONNECTIVITY TO BRT

Year 1:

4.1. Phase 1: Concept Development and Internal Analysis (0-6 months)

4.1.1. Define Need and Corridor Selection (1-2 months)

- Dense, bottom-30% income areas
- High carless rates
- Weak or zero feeder connections to G/J Line BRT corridors
- High concentrations of Hispanic and other disadvantaged communities
- These tracts form the basis for corridor selection and equity justification.

4.1.2. Technical Feasibility Study (2-3 months)

- Need to define route alignments, stop spacing, street width constraints, and headways.
- Fleet options: 30-35 ft battery-electric buses or CNG, where electric readiness is limited.
- Vehicle sizing (mid-size vs micro-bus) is determined based on street width, turning radii, expected demand, and zero-emission suitability
- Capital needs: cost of new buses, shelters, lighting, safe transfer hubs, chargers
- Operational costs from labour and maintenance.
- Conduct early risk screening: safety: potentially lack of pedestrian access, NIMBY concerns: noise pollution concerns, curb use conflicts: will that impact bike lane, outdoor dining, parking.

4.1.3. Internal Metro Review (1 month)

Metro Planning, Operations, the relevant Service Councils (notably Gateway Cities based on empirical analysis), and Accessibility Advisory Committees review alignment and feasibility

The project is formally classified as a Pilot Feeder Programme because introduces new feeder loops, has new mixed fleet, and requires monitoring of ridership, safety, transfer reliability and equity metrics.



Figure 11. Map of Metro Service Council Regions (LA Metro 2020)

4.2. Phase 2: Public Engagement and Governance Approvals (4-6 months)

4.2.1. Early Stakeholder Consultation (1-2 months)

After having consulted the Metro Council to present the feasibility study, it will be necessary that Metro and LADOT engage neighbourhood groups, disability and women's safety advocates, schools, clinics, local businesses, and municipal bus operators through:

- Public workshops (virtual + in-person), surveys, multilingual materials: Spanish,
- Identifying safety fears, refining stop locations, and detecting unintended consequences.

4.2.2. Governance Approvals

Board-level approval is required for new routes, new fleet procurement, and infrastructure modification. Political resistance may delay approval.

Year 2:

4.3. Procurement and Operational Preparation (6-12 months)

4.3.1. Fleet Procurement or Allocation (6-12 months)

- Deciding that Metro either purchase the fleet or do PPP leases for BEBs. Procurement can take 9–18 months for BEBs.
- Then, Infrastructure (charger costs notably) may require city permits, knowing that Coordination with LADOT + Bureau of Street Services is typically slow.

4.3.2. Bus Stop and Streetscape Improvements (3-6 months)

Including shelters, lighting, ADA access, visibility standards, and road markings.

4.3.3. Operator Training, Scheduling, and Run-cutting (2-3 months)

- Operator training on new vehicles and turning paths
- Run-cutting, crew scheduling

- GTFS, NextBus and trip-planner updates

4.4. Pilot Development and Full Operation (3-6 months)

4.4.1. Pilot Launch (3-6 months)

Metro deploys feeder in 3-4 underserved tracts, monitoring:

- on-time performance
- transfer reliability
- ridership and load factors
- equity indicators (looking at who benefits)
- feedback from surveys on security concerns, global perception and on what to improve.
- environmental monitoring: kWh per mile, charger utilisation, avoided emissions vs CNG baseline

4.4.2. Adjustment Period (1-2 months)

Modification of stop locations, spacings, headways, branching, fleet allocation.

4.4.3. Full Implementation (Ongoing)

- End of Pilot feeder programme
- Solution integrated into Metro's Spring/Fall service change cycles.
- Annual performance audits every 6–12 months.

The total implementation duration comes down to 24 to 36 months. This is consistent with Metro's NextGen rollout sequencing, North Hollywood Circulator and Microtransit pilot timelines, bus corridor upgrades under Metro's Transit First / BRT programmes, and typical infrastructure permitting timelines with Los Angeles City departments.

Year 3:

4.5. Pilot Launch Cost Estimation

The table summarises indicative capital (CAPEX) and operating (OPEX) costs for the pilot deployment in 3–4 priority tracts. Estimates cover the small feeder fleet required for the pilot, associated charging or CNG infrastructure, and essential stop and streetscape upgrades at BRT transfer points. CAPEX reflects one-time investments needed to launch the service, while OPEX captures annual labour, maintenance, energy, and service supervision costs. Cost ranges are based on LA Metro procurement data, FTA Low/No Emission Bus benchmarks, and typical US industry averages for shelters, charging equipment, and operations (Metro Board, 2023. FTA, 2025).

Overall, the pilot is financially modest compared with rail or BRT expansion, with total CAPEX in the USD 25–51 million range and annual OPEX estimated at USD 9.6–22.6 million. These values illustrate that a feeder-based intervention is operationally feasible and scalable, even when accounting for the need for ongoing subsidy in low-income areas.

Item	Assumptions	Unit Cost [USD]	Quantity	Total Cost [USD]
Battery-electric Buses (BEBs)	30-35 ft; 2-3 Buses/Route; 3 Routes/Tract	\$0.90M– \$1.10M	24–36 buses	\$22M–\$40M
Charging	Depot Chargers and	\$40k–\$90k per	12–20	\$0.5M–\$1.8M

Infrastructure Bus Stop and Streetscape Upgrades	Power Upgrades Shelters, Lighting, ADA Access; 6-10 Stops/Route	charger \$15k–\$40k per stop	chargers 80–120 stops	\$1.2M–\$4.8M
Pedestrian Access Upgrades	Crosswalks, Curb Ramps, Lighting	Lump sum		\$1M–\$3M
Transfer Hub Improvements	Wayfinding, Lighting, Signage at BRT Interfaces	\$250k–\$500k per hub	4 hubs	\$1M–\$2M
Total CAPEX				\$25M–\$51M
Annual Bus Operations	Labour and Maintenance	\$350k–\$550k per bus/yr	24–36 buses	\$8.4M–\$19.8M
Energy/Fuel	Electricity or CNG	\$20k–\$40k per bus/yr	24–36 buses	\$0.5M–\$1.4M
Control and Scheduling	Dispatch, Supervisors	Lump Sum		\$0.5M–\$1M
Community Engagement (Year 1)	Workshops, Outreach	Lump sum		\$0.2M–\$0.4M
Total OPEX (Annual)				\$9.6M–\$22.6M

Table 1. Indicative Capital and Operating Costs for the Pilot Deployment in 3-4 Priority Tracts.

5. ENVIRONMENTAL SUSTAINABILITY

The proposed feeder bus system is strongly aligned with Los Angeles’ long-term climate, air-quality, and environmental justice goals. LA Metro has committed to achieving a 100% zero-emission bus fleet by 2030 (Rick Jager 2020), and feeder routes present an opportunity to accelerate the deployment of battery-electric buses. From an engineering standpoint, small- and mid-size ARBOC’s battery-electricity buses (30-35 ft) (METRO 2021) are well suited to neighbourhood-scale feeder operations due to lower noise, shorter turning radii, and reduced energy consumption on short-stop spacing routes, which is particularly relevant in neighbourhoods that historically experience higher environmental burdens. If updating to electric infrastructure is not immediately available, CNG buses remain a viable interim option. Metro’s existing fleet composition and depot readiness make a phased electrification strategy more feasible than an immediate full rollout of battery-electric buses. This reflects standard transition practice in North American agencies undergoing fleet electrification.



Figure 12. ARBOC's Battery-Electric, Low-Floor Bus (METRO 2021)

By targeting dense low-car-ownership areas, the feeder system also supports LA's Metro's Moving Beyond Sustainability Plan, which prioritises reducing vehicle miles travelled by improving access to rapid transit. Improving connections to rail and BRT corridors reduces reliance on private vehicles and ride-hailing, enabling more utilisation of existing high-capacity infrastructure, hence increasing sustainable travel.

In the long term, resilience is strengthened by the system's flexibility. The feeder system would become a permanent, scalable layer of LA's transit network, expanding as demand grows and new Metro or BRT stations open. Feeder routes can be scaled or reconfigured as land use evolves, supporting emerging transit-oriented development areas. Environmental monitoring avoided emissions relative to CNG baselines, and charging infrastructure uptime should all be considered in pilot deployment.

Overall, the proposed feeder network advances climate mitigation, environmental justice, and long-term system resilience, positioning it as a critical component of a zero-emission, equitable transit future for Los Angeles.

6. SOCIAL AND EQUITY CONSIDERATIONS

The project does not assume universal demand. In some neighbourhoods, distrust toward new transport interventions persists due to concerns around safety and past policing practices. This is recognised as a risk to project success and highlights the need for community-led engagement rather than top-down implementation. At the same time, improved feeder connectivity offers clear equity benefits. Census tracts with the weakest access to the G and J Line corridors are disproportionately low-income, predominantly Hispanic, and highly car dependent. In several of these areas, carlessness rates exceed 50–60%. This means residents cannot offset limited transit provision through private vehicles. Strengthening feeder routes therefore directly improves mobility for those most reliant on public transport.

Recent evidence indicates that safety concerns are now a primary barrier to transit use in Los Angeles. Post-pandemic Metro surveys show that perceived insecurity on buses and around stations is a key reason some former riders have not returned. Crime data reinforce this trend, with violent crime on public transport and stations increasing by nearly 55% between 2020 and 2023 (Ileana Wachtel 2024). Addressing safety perceptions and building trust through community participation are therefore essential enabling conditions, without which even efficient network designs risk limited uptake.

Feeder-route planning must therefore incorporate meaningful stakeholder engagement and genuine community engagement. Public transport planning literature shows that participatory processes can improve uptake and trust by aligning services with local needs, particularly in areas where security and policing have been contested (Lucas 2012; Golub and Martens 2014). Design measures such as predictable headways, improved lighting, and clear visibility standards further support safer conditions. Cities such as Montréal and Dublin have also adopted evening request-stop policies to improve safety for women and vulnerable passengers (Mass Transit 2022).

From an equity perspective, improved access to BRT enhances access to key destinations for low-income and carless households while supporting inclusive mobility. Ensuring accessibility for disabled passengers and older residents strengthens social value and helps secure political and institutional support from stakeholders. These include LA Metro, LADOT, and local government, increasing the likelihood of sustained investment.

7. ECONOMIC AND INSTITUTIONAL FEASIBILITY

Delivering a strengthened feeder-bus network requires navigating significant financial and institutional constraints. International transport research shows that service reorganisation is among the most cost-effective ways to improve accessibility, because most expenditures relate to labour and operations rather than infrastructure (Vukan R. Vuchic 2007). However, this proposal is not limited to restructuring existing routes but also to implementing new neighbourhood-scale feeder loops, which require fleet expansion. These capital requirements must be planned for.

Operational expansion is also not cost-neutral. Evidence from US transit systems indicates that new local routes in lower-income districts typically require sustained subsidy as ridership in lower-income districts rarely covers operating costs (Taylor and Fink 2003). This strengthens the rationale for public investment, especially given the strong equity and accessibility benefits demonstrated earlier.

Several financing mechanisms exist. Reallocating existing Metro operating budgets offers one pathway, while federal programmes, such as the FTA Low- or No-Emission Bus Program, can support higher capital costs of fleet upgrades and electrification. Public–private partnerships (PPPs) provide another option. Cities like Bogotá have procured large portions of their electric feeder fleets through such arrangements, allowing rapid deployment with limited upfront expenditure for the operator (Samantha Pettigrew et al. 2023). Yet PPPs also present well-documented risks: long term contracts can reduce flexibility, and poorly structured agreements may shift financial risk to the public sector (Siemiatycki 2015). Clear governance and performance metrics are therefore essential.

Importantly, private operators are generally unwilling to serve low-income districts because feeder routes rarely earn enough fare revenue to cover their operating costs. Even when ridership increases after integration with BRT, Bogotá reported 40–60% gains on some feeder services; local circulators typically remain subsidy-dependent due to short route lengths, dispersed travel patterns, and a high share of discounted-fare riders. This does not weaken the case for investment. Other international development projects, most notably from Houston and Bogotá, show that when stable subsidy and effective inter-agency coordination are in place, feeder-route reorganisation can deliver large accessibility improvements at comparatively low cost. This is especially true if the high social returns of connecting carless and disadvantaged residents to major employment and service centres are accounted for in the analysis.

Finally, institutionally, successful delivery requires coordination between LA Metro, LADOT, local councils, transit unions and community organisations. Fragmented governance has historically impeded transit reforms in US cities, often generating duplication or political resistance to route redesign (Hidalgo et al., n.d.).

8. CONCLUSION

Using spatial and demographic analysis, the study identified neighbourhoods where accessibility gaps are most pronounced and where targeted feeder-route interventions would deliver the greatest benefit. The findings suggest that district-level feeders can improve integration between local services and high-capacity corridors, reducing journey time variability and increasing the utilisation of existing BRT infrastructure. While the intervention does require modest capital investment, these costs remain far lower than those of implementing new BRT corridors. From an engineering perspective, the approach is scalable and cost-effective, improving network performance primarily through operational restructuring.

The project also highlights that technical design alone is insufficient for successful outcomes. Safety, accessibility and institutional coordination play a significant role in shaping whether proposed services are used in practice. Addressing these issues alongside operational planning is essential to avoid low

uptake and to ensure that improvements translate into meaningful changes in mobility.

Overall, the proposed feeder-based strategy demonstrates how the interventions can strengthen the efficiency and resilience of a complex urban transport network. By improving local-to-corridor integration, the project provides a practical response to challenges addressed to rapid transit. While grounded in the Los Angeles context, the approach offers lessons for other cities seeking to enhance public transport performance through operationally feasible solutions.

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9.1. APPENDIX

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Table 2. Contribution List

Student	Contribution
22021653	- 1. Introduction - 6. Social and equity considerations - 8. Conclusion - Presentation Slides: 2, 8, 9, 12
22016100	- 3. Literature Review - 5. Environmental Sustainability - Presentation Slides: 4, 5, 10 - Report Formatting
22146303	- 2. Empirical patterns and problem definition - 4. Proposed project: strengthening feeder bus connectivity to BRT - 7. Economic and institutional feasibility - Presentation Slides: 3, 6, 7, 11